

Build, image, containerise

Day 2 - Afternoon

Running scenario: Thara Hospitals patient portal pilot

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CONCEPT 1



Tagging and the ledger

10 min

Three tags on everything, and the ledger first

- Every resource today: `Module=CC, Day=2, Owner=<student id>`
- A tag changes nothing about the resource; it lets you find it and cost it
- The ledger row is written **before** teardown, not after

Once a resource is deleted, the console can no longer tell you its id or its start time.

CONCEPT 2

What you are about to do that is wrong

A leaked access key is how the leaked-link incident could have been much worse.

10 min

Step 10 is the insecure method, deliberately

- You put a long-lived access key on the instance to reach S3
- Day 5 replaces it with an instance role. The contrast is the point
- Fenced in: off today, on for minutes on Day 4, deleted on Day 5

COMMON MISTAKE

Deactivating a key deletes it.

It is only switched off. Day 5 removes it.



A key that never expires, in a text file on a server.

Build, image, containerise



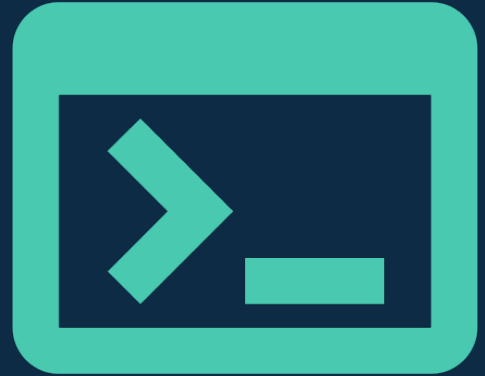
You build:

2

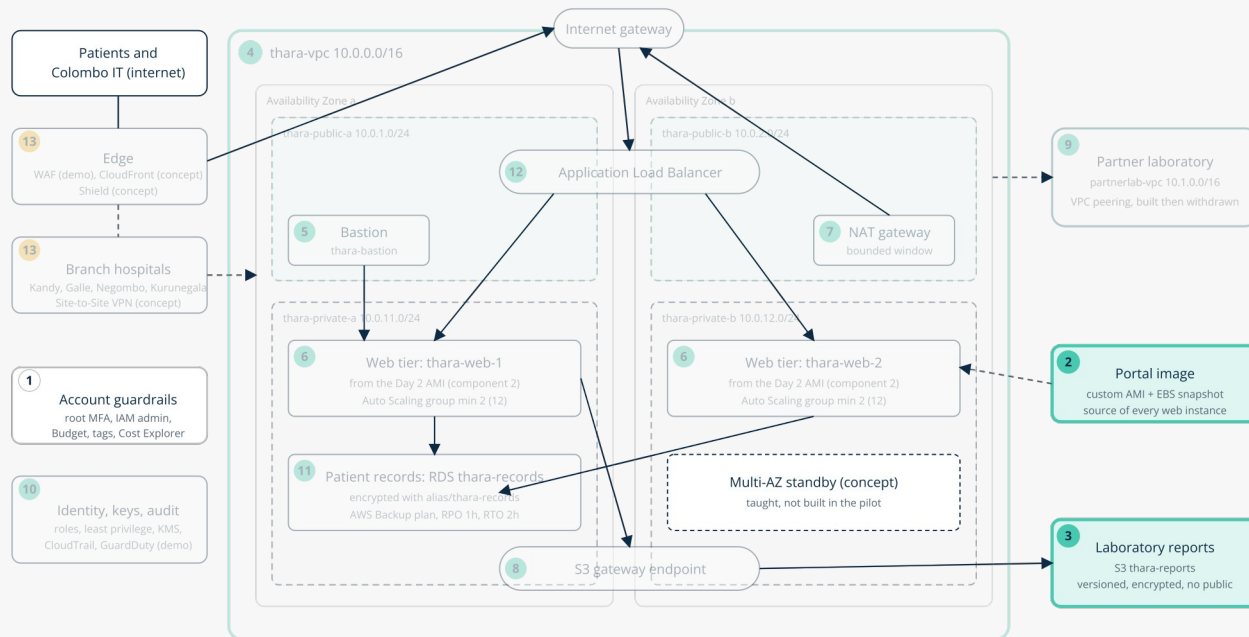
3

Estimated credit use: about USD 0.60

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The Thara architecture



TODAY

- 2** Patient portal web server and image
- 3** Laboratory reports bucket

KEY

- Built today
- Built earlier
- Not yet built

SEGMENT 1 - 30 MIN



Launch and connect

Cost: USD 0.011 per hour plus USD 0.005 per hour for IPv4

Check, then launch the web server



1

Confirm the region is `ap-south-1` and that you are the admin user, not root.

Expected: Mumbai, and `admin-<student id>`, in the top bar.

2

From Explore AWS, launch `thara-web-1`: Amazon Linux, t3.micro, new key pair `thara-key`, default VPC, SSH from My IP, HTTP from the internet, the three tags.

Expected: Running, with 2/2 checks passed after about two minutes.

`thara-key.pem` downloads once. Keep it safe: Day 3 needs it.

The user data for step 2



```
# paste under Advanced details, User data
#!/bin/bash
dnf install -y httpd
echo "<h1>Thara Hospitals portal (pilot)</h1>" \
    > /var/www/html/index.html
systemctl enable --now httpd
```

The script runs once, at first boot, as root. Nobody logs in to run it.

Connect, and load the page



3

Add an inbound SSH rule whose source is the prefix list ending `ec2-instance-connect`. Then Connect, EC2 Instance Connect. Browse to `http://` and the public IPv4 address.

Expected: A terminal as `ec2-user`, and the page Thara Hospitals portal (pilot).

The browser terminal comes from the Instance Connect service, not from your IP.

SEGMENT 2 - 35 MIN



EBS attach, format, snapshot

Cost: 8 GB gp3, under USD 0.01 for the afternoon

Attach a data volume and snapshot it

4

Create an 8 GiB gp3 volume in the instance's zone, tag it, attach it as `/dev/sdf`. Format, mount at `/data`, write `reports-index.txt`.

Expected: df shows 8.0G mounted on /data.

5

Snapshot the volume with the three tags. Write the snapshot id in your ledger.

Expected: The snapshot shows Completed.

```
$ lsblk
$ sudo mkfs -t xfs /dev/nvme1n1
$ sudo mkdir /data && sudo mount /dev/nvme1n1 /data
$ echo "Thara reports index" | sudo tee /data/reports-index.txt
```

SEGMENT 3 - 15 MIN



Create the AMI

Cost: snapshot storage, a few cents a month

Create the image

6

Actions, Image and templates, Create image, named `thara-portal-v1`. Remove the 8 GiB data volume from Instance volumes. Tag image and snapshots together.

Expected: The AMI moves from Pending to Available. Your terminal drops: the instance reboots.

```
# reconnect, then mount the data volume again
$ sudo mount /dev/nvme1n1 /data
$ cat /data/reports-index.txt
```

Leave the data volume in, and every server launched from this image gets an extra disk.

SEGMENT 4 - 35 MIN



Docker contrast

Cost: nil

Run a container beside the native server

7

Install Docker and run nginx mapped to port 8080. Add an inbound rule for TCP 8080 from My IP. Browse to port 80 and to port 8080.

Expected: Port 80: the Thara pilot page. Port 8080: Welcome to nginx!

```
$ sudo dnf install -y docker
$ sudo systemctl enable --now docker
$ sudo docker run -d --name web8080 -p 8080:80 nginx
```

Compare the container with the host

8

Compare processes, memory and the file-system view from the host and from inside the container. Record three differences.

Expected: nginx is a host process; no second OS uses memory; the container cannot see /data.

```
$ ps -ef | grep -E "httpd|nginx"
$ sudo docker top web8080
$ sudo docker stats --no-stream web8080
$ sudo docker exec web8080 cat /etc/os-release
$ sudo docker exec web8080 ls /data
```

SEGMENT 5 - 25 MIN



Reports bucket and the insecure upload

Cost: nil

Create the reports bucket



9

Create `thara-reports-<student id>` in `ap-south-1`. Leave ACLs disabled and Block all public access ticked. Add the three tags.

Expected: Permissions shows Block all public access: On.

Insecure, deliberately: the long-lived key

10

IAM, your admin user, Security credentials, Create access key, Command Line Interface. Download the .csv. Configure the CLI on the instance and upload the file.

Expected: An upload line naming `thara-reports-<student id>`, then the object listed.

```
$ aws configure
# <bucket> is thara-reports-<student id>
$ aws s3 cp /data/reports-index.txt s3://<bucket>/
$ aws s3 ls s3://<bucket>
```

The secret now sits in plain text on the server. Ledger note now; never in a screenshot.

SEGMENT 6 - 20 MIN



Ledger, teardown, evidence

Cost: nil

Prove it, then break it



PROVE IT

- `curl http://localhost` returns the pilot page; `curl http://localhost:8080` returns nginx
- `aws s3 ls` on the bucket, from the instance, lists the object
- The AMI `thara-portal-v1` shows Available

BREAK IT

- Remove the HTTP rule from the security group and reload the page
- Observe: a timeout, while `curl http://localhost` still works
- Restore the rule. Note which layer the failure happened at

Tear down and record



TEARDOWN

- Remove the 8080 rule. Stop `thara-web-1`; do not terminate it
- Detach and delete the 8 GiB volume; keep the snapshot and the AMI
- Deactivate the access key; do not delete it. Keep the bucket

COST

Estimated credit use: about USD 0.60

FOR YOUR EVIDENCE PACK

- Screenshot of the pilot page and the nginx page side by side
- The three recorded VM-versus-container differences
- AMI id and snapshot id; bucket name with Block Public Access shown on
- Ledger row with the note about the access key

Common mistakes



COMMON MISTAKE

The container is inside a separate machine.

It is a process on the same kernel.

COMMON MISTAKE

Deactivating a key deletes it.

It does not. Day 5 removes it.

Before the next session



READ

Amazon VPC User Guide

How Amazon VPC works. Then Wittig and Wittig, the VPC chapter.

SELF-CHECK

CLF-C02 bank

Domain 3 storage set, 10 items.

PQ2 is due.

TASK

Carve 10.0.0.0/16

Four subnets across two AZs, two public and two private. Say why each is which.

On paper. Bring it to Day 3.

Questions & Discussion



Next: the network. You build `thara-vpc` and prove the image inside it.



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